

# Building the Credence Capability

A development log that doubles as a seminar

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# What this document is

A running record of how the Credence teaching app is being extended — written as a seminar rather than as changelog entries, because most of what we learned is worth teaching.

## Three threads run through it:

1. **The Credence framework** — what it does, and the boundary between *realism* and *truth* that makes it work.
2. **The road to CausalMix** — where a hand-rolled conditional VAE runs out of road, argued from defects we actually hit rather than from the abstract.
3. **survcausal-pilot** — the larger project this feeds. Every entry closes by saying what it bought *there*, not just here.

**Companion to:** [tutorial/tutorial.pdf](#) (Module 5, instats SSC 2026).

## **Part I — The framework**

Realism vs. truth; why the split matters.

## **Part II — Tier 0: making the claims true**

Six defects, one of which was found *by* the fix for another.

## **Part III — Toward CausalMix**

Post-hoc repair vs. architectural fix.

## **Part IV — `survcausal-pilot`**

The larger project: what it is, what it still needs, and how this work supplies it.

## **Part V — Tier 1**

$\tau(X)$  heterogeneity, an overlap dial, and factorial scenarios.

## Part I — The framework

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# The problem Credence solves

You have an estimator. Does it work on *your* data? You cannot tell, because the individual causal effect is never observed. So the field validates on simulation — and simulation has its own problem:

- we hand-specify the data-generating process;
- a hand-specified DGP embeds parametric assumptions;
- those assumptions may quietly **favour some estimators over others**.

Credence (Parikh et al., ICML 2022) separates the two things a benchmark needs:

## REALISM

learn  $p(X | Z)$  from real data

## TRUTH

choose  $\tau, \rho, f(X)$  yourself

**The trick:** because *you* chose  $\tau$ , you can score any estimator. Because the covariates came from real data, the benchmark is not a toy.

# The boundary that has to hold

The generator learns  $p(X \mid Z)$  and **never sees the outcome**.

This is not a detail. If the generator knew the treatment effect, then “recovering” that effect downstream would prove nothing — you would be measuring how well information flows through your own pipeline, not whether an estimator works.

In the code, that boundary is a file boundary:

- `cvae_model.py` — realism. No  $Y$ , no  $\tau$ , anywhere in it.
- `dgp.py` — truth. Chooses  $\tau$ ,  $\rho$ ,  $f(X)$ .
- `realism.py` — evidence. Checks the first one actually worked.

**Why it matters:** Keeping realism and truth in separate files makes the boundary auditable. A reviewer can confirm the generator never touches  $Y$  by reading one import list.

## The generative order looks backwards (and is fine)

In the world,  $X$  causes  $Z$ : sicker patients are treated differently. In the app we draw  $Z$  first, then  $X \mid Z$ .

That is not an error.  $p(X, Z) = p(Z) p(X \mid Z)$  is a valid factorization of the same joint distribution — we are *sampling from the joint*, not asserting a causal direction. A conditional VAE finds  $p(X \mid Z)$  far easier to learn than the reverse.

**What it does cost us:** the *strength* of observed confounding is inherited from whatever the real data happened to contain. It is not a dial.

Making it a dial — along with overlap — is exactly what Part V is about.

## Part II — Tier 0: making the claims true

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## Where we started

The app told users its output was *“realistic and near-indistinguishable from the observed sample.”*

It never checked. And underneath the claim were six defects.

| What was wrong                  | Consequence                            |
|---------------------------------|----------------------------------------|
| One gradient step per epoch     | generator barely trained               |
| Decoder round-trip never called | export did not match your schema       |
| $f(X)$ selected columns by name | snake_case covariates silently dropped |
| $f(X)$ averaged on raw scale    | dollars drowned out years              |
| No seed on generation           | nothing reproducible                   |
| No realism check at all         | the headline claim was unfalsifiable   |

**The through-line:** an app that asserts its own quality without evidence is precisely the thing Module E trains participants to catch.

## Defect 1 — fifty gradient steps

The training loop called `optimizer.step()` **once per epoch**, on the whole dataset at once. At the app's default of 50 epochs, that is **50 gradient steps in total**.

**The defect:** The “learned covariate structure” was closer to the untrained prior than to the data — while the interface promised near-indistinguishability.

**Fixed:** minibatching, with  $\lceil n/\text{batch} \rceil$  steps per epoch, a seeded shuffle, and a per-observation ELBO so the reported number means the same thing at any batch size.

Measured, pneumonia cohort, 5 epochs: full-batch = 1 step/epoch, -ELBO 3.62; minibatch(128) = 8 steps/epoch, -ELBO **2.93**.

**Why it matters:** We kept “Full dataset (the old bug)” as a menu option, so the workshop can *reproduce* the failure and watch the realism score degrade — rather than take our word for it.

# The fix that mattered most — measuring realism

Three checks, weakest to strongest, all against **held-out rows the generator never trained on**:

| Check             | Question                            | Catches                    |
|-------------------|-------------------------------------|----------------------------|
| SMD               | are the means close?                | location                   |
| KS distance       | are the <i>distributions</i> close? | spread, skew, modes        |
| Discriminator AUC | can a model tell them apart?        | broken <i>correlations</i> |

**Why all three.** A generator can match every marginal mean and get every shape wrong. It can match every marginal distribution and still destroy the *joint* structure — and only the discriminator, which sees whole rows, can notice that.

**AUC  $\approx 0.5$  is the target.** This is the rare model you want to perform badly. Participants find that disorienting, which makes it stick.

## Why the holdout is not optional

These metrics must compare synthetic data against real rows the generator has **never seen**.

Scored against its own training rows, a flexible generator wins by **memorising**. That is:

- a meaningless result — it says nothing about generalisation; and
- a **privacy problem** — a generator that memorises patients can leak them, and participants upload their own files.

So Step 2 now splits before it trains, and Step 4 compares against the holdout.

**Why it matters:** Privacy metrics are one of the things CausalMix ships and we do not. The holdout is the minimum honest version of that idea.

## Defect 10 — the diagnostic found a bug we had not listed

First run of the new realism check, after 150 epochs of proper training:

**Discriminator AUC = 1.000**

Perfect separation. The obvious reading — “the generator is hopeless” — was wrong.

| Covariate      | Real   | Synthetic                |
|----------------|--------|--------------------------|
| priorPneumonia | {0, 1} | [−0.30, 1.35] continuous |
| priorVaccine   | {0, 1} | [−0.25, 1.35] continuous |

The decoder round-trip only ever handled `object/Categorical` columns. **Binary and integer covariates were exported as continuous relaxations.** The classifier was not detecting anything subtle — it was asking “is this value exactly 0 or 1?”

**Why it matters:** The diagnostic paid for itself on its first run — by finding a defect that was not on the list it was built to check.

# Sampling, not rounding

Fixing it raises a genuine statistical choice. Given decoder output implying  $P = (0.62, 0.31, 0.07)$  across three categories:

## argmax / threshold

Always take the most likely.

- every row  $\rightarrow$  category 1
- rare categories **vanish**
- synthetic data less variable than real

## sample

Draw with those probabilities.

- frequencies come out right
- marginal distribution preserved
- costs nothing

What we need to preserve is the **marginal distribution**, not the most likely value of any single row. So we sample — categorically for categories, Bernoulli for binaries, round-and-clip for integers.

Measured: with a true rare-level share of 15%, sampling returned **16.7%**; argmax returned **0%**.

## Defect 4 and 5 — what $f(X)$ was

$f(X)$  is the prognostic index: how covariates move  $Y(0)$ . It *is* the observed-confounding mechanism. It was one line:

```
numeric = [i for i, name in enumerate(names) if '_' not in name]  
hX = X[:, numeric].mean(axis=1)
```

### Two independent defects.

1. **Selection by name** — keeps columns whose name has no underscore. Meant to skip one-hot dummies; also drops any covariate named `prior_vaccine`. Still *exported*, so it looked like a confounder and influenced nothing.
2. **Raw-scale averaging** — an unweighted mean across units is decided by whichever column has the biggest numbers.

**Why it matters:** Both built-in datasets use names without underscores — so it survived. The first participant to upload their own file would have hit it.

## ... and what $f(X)$ is now

Every encoded column participates — selection by *name* is gone. Each is standardised on the **training** statistics, weighted, and rescaled to a stated magnitude:

$$f(X) = c \cdot \sigma_Y \cdot \frac{\sum_j w_j z_j}{s}, \quad z_j = \frac{x_j - \bar{x}_j}{\text{sd}_j}$$

- **unit-invariant** — rescale earnings and  $f(X)$  does not move;
- **weights visible in the UI** — weight 0 means cannot confound;
- **magnitude interpretable** —  $c$  is in outcome standard deviations.

**Why it matters:** Direction is a stated choice ( $w$ ); magnitude is a stated parameter ( $c$ ). Turning  $c$  into a dial is Tier 1.

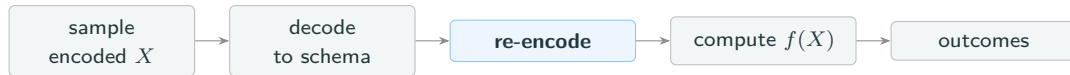


# The property that makes a benchmark usable

**The exported covariates must fully determine the potential outcomes** (up to  $U$  and noise).

If the outcome secretly depends on something we did not export, then *no* analyst can adjust for it, *every* estimator looks biased — and the workshop teaches a false lesson about the methods instead of a true one about the data.

This forced a step the old pipeline did not have:



Without the re-encode, a categorical's *exported* value only loosely tracks the value that actually drove the outcome.

## Two seeds, on purpose

Generation used unseeded draws: every click produced different data and nothing was reproducible.

The fix is not one seed but **two** — training and generation, kept apart:

- redraw a cohort *without* retraining the model;
- retrain *without* moving the cohort;
- so you always know which change caused what you are looking at.

This mirrors `survcausal-pilot`'s own convention, which seeds the cohort draw and the estimator fits from **disjoint streams** for exactly this reason (`sseed` / `eseed` in `run_pilot.R`).

**Why it matters:** Attribution. The same principle as the pilot's factorial design: move one thing at a time, so any difference has a single cause.

## The gate: what we refuse to ship without

Ten checks, runnable with `python3 tests/test_tier0.py` — no pytest, so a participant can run it on a workshop laptop. The one that matters most:

**The truth gate.** With  $\rho = 0$  the only path from  $Z$  to  $Y$  other than the effect runs through the exported covariates. A correctly specified adjustment *must* recover  $\tau$ .

|                  | estimate | bias   |
|------------------|----------|--------|
| true $\tau$      | -5.000   | —      |
| naive difference | -4.349   | +0.651 |
| g-computation    | -4.977   | +0.023 |

(tolerance  $\pm 0.130$ , four Monte Carlo standard errors)

**Its complement.** At  $\rho = 0.8$ : adjusting for  $X$  leaves bias +0.284; adding  $U$  brings it to -0.015.

**Why it matters:** If the truth gate fails, the generator is wrong and nothing downstream can be trusted.

## Tier 0 — result

| Discriminator AUC                           | before | after        |
|---------------------------------------------|--------|--------------|
| undertrained generator (1 full-batch epoch) | 0.852  |              |
| properly trained generator                  | 1.000  | <b>0.602</b> |
| app flow, pneumonia cohort                  | —      | <b>0.579</b> |

The 1.000 → 0.602 column is the type-aware decoder (defect 10). The 0.852 figure is what an undertrained generator still scores *after* that fix — which is why both numbers are reported: the **improvement** is the teaching artifact, not the absolute level.

**Verdict at 0.579:** *“Indistinguishable — the discriminator is near chance.”* The claim the app was already making is now one the app can defend.

## Part III — Toward CausalMix

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# What CausalMix is

**CausalMix** — Zhang, **Parikh**, Naimi, Nabi, Kim, Lash (arXiv:2603.03587, March 2026).

A conditional VAE with:

- a **mixture-of-Gaussians latent prior** — real covariates are multimodal and clustered; a single Gaussian smooths that away;
- **data-type-specific decoders** for continuous, binary and categorical columns;
- an **overlap regularizer** shaping the propensity-score distribution;
- overlap, confounding strength and effect heterogeneity varied **independently at design time**.

**The decisive fact:** Harsh Parikh is first author of Credence *and* an author of CausalMix. This is not a competing school — it is Credence's own lineage, four years on. Adopting it is a continuation, not a retraction.

## Defect 10, revisited — the case for typed decoders

Our fix for binary and integer columns is **post-hoc repair**: the network is trained as if every column were continuous, and we discretize its output afterwards.

### What we did

- MSE loss on every column
- Bernoulli-sample binaries after decoding
- round-and-clip integers after decoding

Works. AUC 1.000  $\rightarrow$  0.602.

The architectural fix is to give the model a per-column likelihood and decode each type natively — which is exactly what CausalMix does.

### What it cannot fix

- the *training signal* still treats a 0/1 flag as a continuous target
- so the *joint* structure between a binary and a continuous covariate is learned only approximately

**Why it matters:** We can now argue for Tier 2 with a *measured* number rather than an appeal to authority. That is a better reason to adopt something.

## Caveats we are not going to forget

1. **Maturity.** March 2026 preprint, no visible CI, not on PyPI, effectively one maintainer. Fine for teaching; pin a commit and validate before any number reaches a paper.
2. **Deployment weight.** shinyapps.io already strains with torch. Mitigation — correct regardless of backend — **ship pre-fit generators; let the app only sample.**
3. **No survival outcomes.** Neither Credence nor CausalMix does time-to-event. For survcausal-pilot that is an integration, not a swap.
4. **The parametric assumption moves; it does not vanish.** Realism is bought for  $p(X, Z)$  only.  $f(X)$  and  $\tau$  are still ours.
5. **Self-flattery risk.** A generator fit on the same data used for the benchmark can favour estimators whose inductive bias matches the generator — a VAE's smooth latent structure flatters smooth-function learners.

**Say caveat 5 out loud in the workshop.** Credence's entire pitch is that hand-specified DGPs may inadvertently favour certain estimators. The same critique applies to us.



## Part IV — survcausal-pilot

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# The larger project

survcausal-pilot — a pilot for a causal-estimator extension to Diana Shamsutdinova's survcompare package. *Shamsutdinova–Shuryak–Yang collaboration*.

## Two questions, deliberately not conflated:

### What will happen?

Prognosis. Ranks patients by risk. Scored on discrimination and calibration.

A patient can be high-risk and gain little; lower-risk and gain a lot. A purely predictive model conflates the two.

### What should we do?

Benefit. Targets the treatment effect. Scored on RMSE against a known  $ATE(t)$ .

**The causal question needs known truth** — which only simulation can give. That is where this work connects.

## The pilot's headline result

A  $2 \times 2$  factorial Monte Carlo (50 replicates/cell), crossing confounding shape with hazard shape *at a single fixed effect size*, so any difference has one cause.

| relative RMSE      | Cox-PH       | RSF   | CSF   | CAST (spline) |
|--------------------|--------------|-------|-------|---------------|
| Linear / PH        | <b>0.087</b> | 0.365 | 0.230 | 0.224         |
| Linear / non-PH    | 0.897        | 0.611 | 0.358 | <b>0.352</b>  |
| Nonlinear / PH     | 0.466        | 0.452 | 0.186 | 0.180         |
| Nonlinear / non-PH | 1.148        | 0.764 | 0.250 | <b>0.239</b>  |

Under simple confounding with proportional hazards a correctly specified Cox model wins outright and causal forests add nothing. Break *either* assumption and the parametric methods collapse while CSF and CAST hold.

**The lesson:** the right method is data-dependent. A benchmark reporting one regime would mislead.

# What the pilot says it still needs

From the pilot's own vignette and README — these are not our inventions:

- *“a data-generating process with genuine **covariate-driven effect modification**”* — named as a natural extension;
- *“A future data-generating process can add genuine covariate-level heterogeneity; the machinery here would estimate it unchanged”;*
- *“**Causal scoring needs known truth.** RMSE against the true  $ATE(t)$  is only possible in simulation, which is why the package ships simulation DGPs with a treatment arm and oracle effect.”*

So the pilot's causal-validation story rests entirely on **simulated DGPs with oracle truth** — and its one soft spot is that those DGPs are hand-specified, in exactly the manner Credence was written to criticise.

**A reviewer will ask.**

## Connection 1 — $\tau(X)$ is the pilot's named gap

The pilot's risk/benefit quadrants (Figure 3) make their point via a *non-monotone* relation between risk and benefit, because the treatment hazard ratio depends only on time — there is **no covariate-driven effect modification** in the cohort. The vignette concedes this twice.

Tier 1's  $\tau(X)$  knob is a prototype of exactly that missing DGP feature.

|                 | app (Python) | pilot (R)           |
|-----------------|--------------|---------------------|
| cost of one run | seconds      | 1–1.5 hours         |
| feedback loop   | immediate    | overnight           |
| stakes          | teaching     | publishable numbers |

**Why it matters:** Prototype  $\tau(X)$  where a run costs seconds; port it where a run costs ninety minutes. The app is the cheap end of the same question.

## Connection 2 — the overlap dial feeds a dormant diagnostic

The pilot *already computes* positivity diagnostics: `pct_extreme` and `pct_clipped` in `run_causal()`, carried through into the summary table.

But nothing in the study ever **varies** overlap. So across all four design cells those columns read:

0.00000, 0.00000, 0.00036, 0.00047

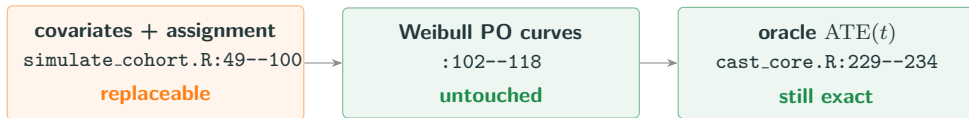
A diagnostic that never moves is a diagnostic that teaches nothing.

Tier 1's overlap slider turns a passive column into a **third design lever** — and it is the axis most likely to separate doubly-robust CSF from the plug-in learners in a way non-proportionality alone does not.

**Why it matters:** Positivity is a first-class identification assumption in Modules C and D. No current teaching asset lets anyone *move* it.

## Connection 3 — the oracle survives semi-synthetic covariates

The strategic prize: anchor the pilot's covariates on a *real* cohort while keeping the known  $\text{ATE}(t)$ . The seam already exists.



**Why the oracle survives.** `true_survprob_ate()` computes  $\tau(t)$  by averaging  $S_1 - S_0$  over the covariate sample. It never assumes the covariates came from a known parametric law — only that the potential-outcome curves are known for each row.

**So a semi-synthetic covariate sample still yields an exactly computable oracle.** That is what makes this viable rather than merely appealing.

# The honest limit

If we anchor covariates on a real cohort, the survival mechanism — the linear predictor `lp_surv` and the treatment model — is **still a hand-specified function** of those covariates.

Realism is bought for  $p(X, W)$ . It is not bought for the outcome model.

This must be stated in the pilot's README if that work ships. Otherwise the pilot would over-claim in precisely the way this dev log spent Part II faulting the app for.

**And one more, for the same reason:** the pilot's data note currently reads *“All cohorts are simulated from a known generative model. No patient data are used or distributed.”* That sentence becomes false the moment a real cohort is touched — and must be rewritten *before*, not after.



## Part V — Tier 1

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## Tier 1 — what we are building

Tier 0 made the app's existing claims true. Tier 1 adds the capability that was genuinely missing — and it is the part that maps onto `survcausal-pilot`.

1.  $\tau(X)$  — **heterogeneous effects**. Constant  $\tau$  is why Step 4 cannot currently tell TMLE, AIPW and g-computation apart: they all recover the same number. With  $\tau(X)$  the comparison acquires teeth.
2. **Overlap as its own dial**, separate from  $\rho$ . Hold  $\tau$  and confounding fixed, degrade overlap, watch estimators fail.
3. **Factorial scenario grid**. Cross  $\{\text{overlap}\} \times \{\text{confounding}\} \times \{\text{heterogeneity}\}$  and summarise in one table — mirroring `run_pilot.R`'s design so that one cause moves at a time.

*All three now built; twelve tests green.*

## $\tau(X)$ — and the property we designed around

$$\tau(X) = \tau_0 + \kappa \cdot \sigma_Y \cdot m(X)$$

$m(X)$  is a *centred* covariate index, so  $\mathbb{E}[\tau(X)] = \tau_0$ .

**Turning heterogeneity on does not move the average effect.** The number the user typed stays the ATE; only the spread changes.

That is deliberate. Without it, a participant raising  $\kappa$  would watch the ATE drift and could not tell whether the estimator had broken or the target had simply moved — the same attribution principle as the two seeds, and as the factorial.

Measured:  $\kappa = 0$  gives ATE  $-5.000$ ,  $\text{sd}(\tau) = 0.000$ ;  $\kappa = 1$  gives ATE  $-5.000$ ,  $\text{sd}(\tau) = 0.852$ .

**Why it matters:** This is the pilot's named v2 gap. `survcausal-pilot`'s vignette concedes twice that its cohort has no covariate-driven effect modification; prototyping it here costs seconds per run instead of ninety minutes.

## Why constant $\tau$ was holding the app back

With a constant effect, g-computation, IPW, AIPW and TMLE all recover the *same number*. A participant comparing them learns only that they agree — the problem is not hard enough to separate them.

It also makes the clinically important question unaskable: *does this work?* versus *does this work for this patient?* Only the second requires  $\tau$  to vary.

With  $\tau(X)$  live there is finally something for a CATE estimator to find. An interacted outcome model recovers the per-patient effect at  $r = 1.000$  on our own DGP — which is the right answer, since that model is correctly specified here. The interesting version of this question starts when it is not.

**Why it matters:** This is the same quantity the pilot's risk/benefit quadrants are built from. The pilot can already *estimate* a CATE; what it lacks is a DGP where the true CATE varies.

# The overlap dial — and the trade it forces

## Learned (Tier 0)

draw  $Z$ , then  $X \mid Z$

- confounding inherited from real data
- realistic
- not adjustable

## Designed (Tier 1)

draw  $X$ , then  $Z \mid X$

- $\text{logit } e = \alpha + \lambda m(X) + \rho U$
- $\lambda$  is a dial
- arm structure is ours, not the data's

**You cannot have both from a conditional VAE.**  $p(X \mid Z)$  and a controllable  $e(X)$  pull in opposite directions.

Getting realistic covariates *and* a designed propensity out of one model is exactly what CausalMix's overlap regulariser is for — the second concrete, measured argument for Tier 2.

## What $\lambda$ actually does

| $\lambda$ | propensity range | % extreme | reading                                   |
|-----------|------------------|-----------|-------------------------------------------|
| 0         | [0.565, 0.565]   | 0.0%      | randomised trial on realistic covariates  |
| 1         | [0.136, 0.895]   | 0.0%      | real confounding, comfortable overlap     |
| 3         | [0.003, 0.999]   | 33.6%     | confounding <i>and</i> positivity failure |

Note the middle row. **Confounding and positivity are separate failures.** At  $\lambda = 1$  the naive estimate is badly biased (+0.68) while every propensity stays comfortably interior — exactly the setting where adjustment is supposed to work, and does.

A dial that conflated the two would teach the wrong lesson. This one does not, and the test suite asserts that it does not.

**Why it matters:** Positivity is a first-class identification assumption in Modules C and D. Until now no teaching asset let anyone *move* it.

# The factorial grid

Three axes crossed, everything else held fixed — generator, covariates,  $\tau_0$ ,  $n$ . Twelve cells. Selected rows:

| overlap    | effect   | hidden  | naive bias | g-comp bias | % extreme |
|------------|----------|---------|------------|-------------|-----------|
| randomised | constant | none    | −0.013     | −0.030      | 0.0%      |
| randomised | varies   | none    | −0.086     | −0.047      | 0.0%      |
| mild       | constant | none    | +0.684     | −0.042      | 0.0%      |
| mild       | varies   | present | +0.994     | +0.072      | 0.2%      |
| poor       | constant | none    | +1.166     | +0.037      | 33.6%     |
| poor       | varies   | present | +1.900     | +0.080      | 35.3%     |

Each axis moves something the others do not — positivity tracks  $\lambda$  alone, effect spread tracks  $\kappa$  alone, residual-bias-after-adjustment tracks  $\rho$  alone. **That separation is asserted by the test suite**, because a factorial whose axes move together is not a factorial.

**Why it matters:** Directly mirrors `run_pilot.R`'s  $2 \times 2$  at fixed effect size. Same discipline, three axes instead of two, seconds instead of hours.

## A limitation the grid exposed

Building the dial surfaced something about the generator.

The prognostic index has **unit** standard deviation on real training data by construction. On synthetic covariates it comes out at **0.852**.

The CVAE, trained with a squared-error reconstruction loss, systematically **shrinks variance** — visible too in the raw ranges: real age spans [7, 116], synthetic [36, 98].

Practical consequence: the overlap dial runs about **15% weak** —  $\lambda = 3$  behaves like roughly  $\lambda = 2.6$ . This is now a tripwire test rather than a footnote: if the ratio collapses further, the grid's axis labels stop meaning what they say.

**Why it matters:** Third measured argument for Tier 2: a mixture-of-Gaussians prior exists precisely to stop a single-Gaussian latent from collapsing the spread of the data it is modelling.



# Where Tier 1 leaves us

## The app can now

- vary the effect across patients
- dial positivity independently of confounding
- run a 12-cell attributable factorial
- score two estimators against known truth in every cell

## Three measured arguments for Tier 2

1. typed decoders — defect 10
2. overlap regulariser — the trade on the previous slide
3. mixture prior — variance shrinkage

None of these three was an appeal to authority. Each came out of a number we watched go wrong.

**Next:** Tier 2 puts the generator behind an interface and ships pre-fit artifacts; Tier 3 is the `survcausal-pilot` integration — real covariates, oracle  $ATE(t)$  intact.

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